H524 Introduction to Biostatistics

Final Project: Learn and Apply Statistics with AI

Fall 2025

Dr. John Molitor

📋 Project Overview

Use AI to teach yourself one statistical method we learned this semester — and apply it to a real question. You'll then teach the class what you learned.

Your Task

Teams of 4-5 students will:

1. Choose a core stats concept (e.g., chi-square test, correlation, regression, t-test, ANOVA, bootstrap, etc.)
2. Learn and apply it using AI tools
3. Compare at least two AI platforms (demo both chat and code)
4. Show what you learned, what worked, what didn't, and how you verified the AI's answers

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Teams

Size: 4-5 students per team

Formation: Self-select teams by Week 4. Post team composition on Canvas discussion board.

Roles: While all team members contribute to all aspects, designate: - **Project Coordinator:** Manage timeline and submissions - **AI Tool Lead:** Coordinate AI platform testing and documentation - **Data/Code Lead:** Ensure reproducibility and code quality - **Presentation Lead:** Coordinate slide development and presentation flow - **Documentation Lead:** Compile prompt logs and write-up

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Statistical Concepts to Choose From

Select ONE of the following core methods from H524:

Option 1: Chi-Square Test

Test for independence or goodness-of-fit with categorical data.

Example Topics: - Is music genre independent of mood tag? - Do voting patterns differ by age group? - Is disease prevalence independent of geographic region?

Option 2: Correlation Analysis

Examine linear relationships between continuous variables.

Example Topics: - Relationship between study hours and exam scores - Air quality index vs respiratory hospital admissions - Social media usage and self-reported well-being

Option 3: Linear Regression

Model continuous outcomes with one or more predictors.

Example Topics: - Do housing prices depend on square footage and location? - Predicting blood pressure from age and BMI - Effect of exercise duration on weight loss

Option 4: t-Test (Independent or Paired)

Compare means between two groups.

Example Topics: - Do organic apples cost more than conventional ones? - Sleep quality before vs after intervention - Math scores: traditional vs flipped classroom

Option 5: ANOVA

Compare means across three or more groups.

Example Topics: - Pain scores across three medication types - Plant growth under different fertilizer treatments - Test anxiety levels across different study methods

Option 6: Bootstrap Confidence Intervals

Use resampling to estimate parameter uncertainty.

Example Topics: - What’s a 95% CI for average delivery time? - Bootstrap CI for correlation coefficient - Median household income CI for small samples

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AI Tool Requirements

You must compare at least 2 of these: - ChatGPT (with or without code interpreter) - Anthropic Claude - Google Gemini - Microsoft Copilot - Other AI tools (with instructor approval)

What to demonstrate: - Show prompts you used with each AI - Compare how each AI explained or solved your problem - Explain what you trusted, corrected, or changed - Demonstrate both chat-based help AND code generation (where applicable)

Timeline

Week	Milestone
Week 4	Teams formed, posted on Canvas
Week 5	**150-word proposal due Sunday 11:59 PM**
Week 6-7	In-class proposal discussions and feedback
Week 8-9	Work on project (office hours available)
Week 10	**Presentations + all materials due**

What to Turn In

1. Presentation (10 minutes total)

Use slides to cover:

Your Question (1-2 min) - What did you want to find out? - Why is it interesting or relevant?

Your Data (1 min) - Where it came from and what's in it - Sample size, variables, any preprocessing

Your Method (2 min) - Which stats concept you chose and why - Brief explanation of the method

Your AI Process (3-4 min) - Show the prompts or examples you tried - Compare how at least 2 AIs explained or solved it - Explain what you trusted, corrected, or changed - Demo both chat assistance and code generation

Your Results (2 min) - Key outputs, graphs, and interpretation - Statistical significance and practical meaning

Takeaways (1 min) - What classmates should learn from your process - Recommendations for using AI in statistics

2. Short Write-Up (2 pages max)

Include: - Which AI tools you used and why - What each AI did well/poorly - One workflow or strategy you'd recommend for others - Any code, screenshots, or outputs that show your steps

3. Prompt Log + Data/Code

Attach: - Prompts and short notes (copy/paste or screenshots) - Dataset (or link to data source) - Code or notebook (if applicable) with clear comments - 2-3 sentences on each teammate's role

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Grading (100 points total)

Category	Points	Description
Question & Data	10	Clear, relevant, ethical use of data
Method & Accuracy	25	Correct test/model, assumptions checked
AI Comparison	25	2+ AIs, transparent logs, clear reflection
Interpretation	15	Results explained clearly and correctly
Reproducibility	15	Code/data provided, process repeatable
Presentation	10	Organized, engaging, within time
Total	**100**	

Deductions: - -5 pts for going over time or missing files - -20 pts if fewer than 2 AIs are compared

Presentation Signup

30 students → roughly 6-8 groups → presented over 2 class sessions in Week 10

Each slot = 10 min presentation + 2 min Q&A

Sign up on the shared sheet (first come, first served). Swaps allowed with instructor approval.

Session	Time	Group/Topic
Session A	00:00-00:12	
Session A	00:12-00:24	
Session A	00:24-00:36	
Session A	00:36-00:48	
Session A	00:48-01:00	
Session A	01:00-01:12	
Session B	00:00-00:12	
Session B	00:12-00:24	
Session B	00:24-00:36	
Session B	00:36-00:48	
Session B	00:48-01:00	
Session B	01:00-01:12	

See separate signup sheet for detailed scheduling.

Tips for Success

1. **Start Early:** Don't wait until Week 9 to begin
2. **Document as You Go:** Save all prompts and AI responses immediately
3. **Verify Everything:** Never trust AI output without checking
4. **Practice Your Presentation:** Time it and rehearse transitions
5. **Use Simple Data:** Don't choose overly complex datasets
6. **Ask for Help:** Use office hours if you're stuck
7. **Be Honest:** Show what didn't work, not just successes

Example Project Flow

Week 5: Proposal Example

"Our team will investigate whether there is a relationship between daily screen time and sleep quality among college students using correlation analysis. We will collect or use existing survey data with variables for hours of screen time and self-reported sleep scores. We plan to use ChatGPT and Claude to help us understand correlation assumptions, generate R code, and interpret results. Team members: Alex (coordinator), Jordan (AI tools), Sam (data/code), Taylor (presentation), Morgan (documentation)."

Week 6-7: What You're Doing

- Finding or collecting your data
- Testing different AI prompts
- Learning about your statistical method from multiple sources
- Starting to code and analyze

Week 10: What You're Showing

- Side-by-side comparison of AI responses
- Your final analysis with interpretation
- Lessons learned about AI strengths and weaknesses
- Recommendations for your classmates

Academic Integrity

Important: - Using AI is required and encouraged for this project - You must be transparent about what AI generated vs what you created - All AI usage must be documented in your prompt log - You are responsible for verifying all AI outputs for accuracy - Each team member must contribute meaningfully - Data fabrication or plagiarism will result in project failure

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Resources

Getting Started with AI Tools

- ChatGPT: https://chat.openai.com
- Claude: https://claude.ai
- Gemini: https://gemini.google.com
- Microsoft Copilot: https://copilot.microsoft.com

Finding Data

See `dataset_suggestions.md` for curated list of accessible datasets.

R Resources

- Course materials on Canvas
- R Documentation: https://www.rdocumentation.org
- Quick-R: https://www.statmethods.net

Statistical Concepts Review

- Course textbook chapters
- Khan Academy Statistics
- Course lecture slides and recordings

Frequently Asked Questions

Q: Can we use a statistical method not listed?

A: Check with the instructor first. It should be a core H524 concept.

Q: What if the AIs give different answers?

A: Perfect! That's exactly what you should discuss in your presentation.

Q: How technical should the write-up be?

A: Focus on the AI comparison and learning process, not just the statistics.

Q: Can we use our own data?

A: Yes, as long as it's appropriate and you have permission to use it.

Q: What if we can't get the analysis to work?

A: Document what went wrong and what you learned. Troubleshooting is valuable.

Q: Do we all present together?

A: Yes, but divide the presentation so everyone speaks roughly equally.

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